# Group and solo flights

Technical report on the FFVL cross-country archive

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**Question.** Do paragliders move differently when they fly in a group, and is it the group or the ground, day and pilots? **Data.** FFVL archive, 154 598 paraglider flights, 38.1-min contacts grouped by a null model (21.7% fly in a group). **Method.** Group time (G), solo flights (S) and the same flights alone (A) compared over *every lag* of the TA-MSD and on phase statistics, inside strata (zone, month, hour, site-day, flight) with 5–95% block-bootstrap bands.

**Findings.**

- **Straighter progress over 1000 s in a group:** $\Delta H_{\mathrm{loc}}(1000) = +0.055$ raw, $+0.037$ within the flight, $+0.043$ at the same site-day and hour (selection only $+0.003$). The gap opens above 150 s; nothing within one glide or climb.
- **It is not the zone:** reweighting solo flights to the cells, months and hours where groups fly removes at most 19% of the gap; $H$: G 0.916, S 0.899 $\rightarrow$ 0.904 at the same site-day and hour.
- **More time climbing, more successful searches, faster and flatter glides, higher climb rate and cross-country speed** once stratified (the pooled readings of climb rate and speed reversed: composition).
- **Size:** most of the effect is there with 2 pilots; at the same place and hour only 7+ stands apart.
- **Mechanisms:** groups form and break in every phase; some pilots lead persistently; groups fly around midday.

**Limits.** Strata fix pilot, day, site, hour and zone, not the decision to fly together; contact = shared air within sight; phases on 1 Hz flights only.

# 1 Setup

## 1.1 Data and grouping

- **Flights.** 156 391 paraglider and 6 094 hang-glider flights on 6 209 days, cleaned tracks of the old repository (soaring, tag `legacy-final`). Each flight is analysed after its *departure* (first fix 3 km from the takeoff); before it, the pilot is in the *local phase* and counts as neither solo nor group. Analyses use loggers of at most 10 s; phases (climb, search, transition=glide) are Vilpellet's runs and exist on 1 Hz flights only (70% of the paragliders).
- **Contact.** $i$ is in contact with $j$ at $t$ if $j$, at some $t'$ with $|t' - t| \leq \delta = 2$ min, was within $\varepsilon_h = 500$ m horizontally and $\varepsilon_z = 300$ m vertically of $i$; when both glide, their headings differ by less than 45°. The delay accounts for pilots who follow the same route.
- **Pair interval.** A contact lasting at least $\tau^* = 38.1$ min, after joining gaps shorter than 2 min.
- **Configuration.** A fixed set of at least two flights that is one connected component of the pair intervals over a maximal interval; anyone joining or leaving starts a new one. **"$n$ pilots" always means the size of the configuration at that instant.**
- **Episode.** A chain of configurations without break, each starting when the previous ends and sharing a member. **Block.** The flights tied by sharing episodes: the unit resampled for the error bars.

| Parameter | Value | How it was set |
|---|---|---|
| $\delta$, largest delay | 2 min | *measured*: ratio of real to chance correspondences is 3.7 at 2 min (Fig. 1, top) |
| $q$, false discovery rate | 0.1 | choice: at most one contact in ten is chance |
| $\tau^*$, shortest contact | 38.1 min | *computed* from $q$ and a null model (Fig. 1, bottom) |
| $\varepsilon_h$, $\varepsilon_z$, $g$ | 500 m, 300 m, 2 min | definitions (shared thermals within sight); no plateau on the data |
| $R_{\mathrm{dep}}$ | 3 km | contacts at the takeoff say nothing |

**Table 1:** Parameters of the grouping.

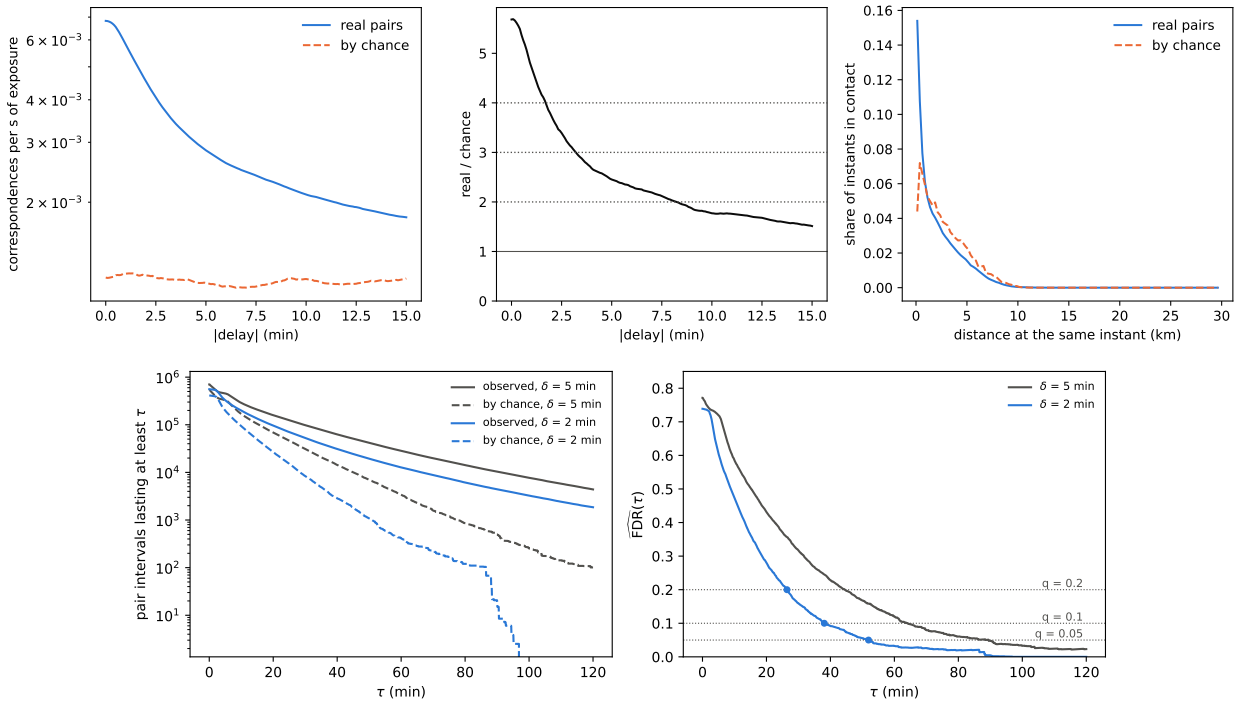



**Figure 1:** How the thresholds were set (tuning note). Top: correspondences per second at each delay, real pairs against chance (left), their ratio (middle), distance at the same instant (right). Bottom: real pair intervals lasting at least $\tau$ and those given by chance (left); estimated false discovery rate (right), with $\tau^*$ for $q = 0.05, 0.1, 0.2$.

**Null model.** Pairs of flights of one day with takeoffs within 10 km (at most 200 per day) are taken, and one flight of each pair is replaced by its own track shifted in time (at least $2\delta + 5$ s, so it cannot reproduce the real contact). Route, thermals and style are kept, the interaction is not. The contacts of the copies are chance: scaled to the flying time of the real pairs, day by day, they give $\widehat{\mathrm{FDR}}(\tau)$; $\tau^*$ is where it falls to $q$. A contact is *shared air within sight*, not proof of interaction: pilots who use the same thermals at the same time without following each other survive the null model.

## 1.2 Populations, strata and statistics

- **S** solo flights; **G** group time (stretches in a configuration); **A** the *same flights alone* (stretches of group flights outside any configuration), the control with the same pilots, gliders, loggers and days; $G_n$ the group time in configurations of $n$ pilots.
- **TA-MSD.** Positions on a 10 s grid of the flight clock; per flight $\overline{\delta^2}(\tau) = \sum \|\mathbf{r}(t+\tau) - \mathbf{r}(t)\|^2 / \#\text{origins}$ pooled over its stretches, ensemble MSD = mean over flights (equal weight); stretches of at least 1250 s, 21 lags from 10 to 1000 s. $H$: half the slope of log MSD against $\log \tau$ over all lags ($H = 1$ straight line). $H_{\mathrm{loc}}(\tau)$: the same slope on the lags within $\pm 0.25$ decades of $\tau$ (Appendix A).
- **Phase statistics.** Ratios of sums over runs (time in climb, searches that end in a climb, climb rate, glide speed, glide ratio; cross-country speed over 30 min). Five of twelve are drawn: the others are explained by them (Appendix C).
- **Strata.** *Zone*: cell of 25 km of the middle of a piece of track; *site-day*: flights of one day with takeoffs within 3 km; *hour*: local hour.
- **Error bars** are 5–95% bootstrap bands resampling whole *blocks* (never flights) and recomputing the statistic: blocks of the note for pooled statistics (1000 draws); blocks joining episodes and site-days for stratified ones (200 draws, same draws for all). They measure block sampling only, not the choice of $q$ or the tolerances, and two bands that do not overlap are not a test; differences are computed draw by draw.

**What the grouping gives.** Of the 154 598 paraglider flights with a logger of at most 10 s, 33 501 (21.7%) fly in a group at some point; of the 6 039 hang-glider flights, 418 (6.9%). Group time is 12.8% of all paraglider flight time after the departure, and on average 45% of the time of a group flight. The 36 575 pair intervals last 40 to 96 min (10th to 90th percentile; median 52 min) and form 40 051 configurations in 17 169 episodes. Half of the episodes have two flights, 1% have more than 9, the largest has 53. By the size of the configuration, the paraglider group time is: 2 pilots 64%, 3 pilots 19%, 4 pilots 8%, 5-6 pilots 6%, 7+ pilots 3% (the largest configuration has 27 pilots). In contact, one pilot is behind the other by a median of 20 s (75% within 45 s).

## 2 Group or zone? Where the difference comes from

**Question.** Group time is more ballistic than solo flights at long lags. Is this the group, or the ground where groups fly (and where solo pilots would also progress in a straight line)? **Method.** Each piece of track is assigned to a *stratum* (zone = cell of 25 km of its middle; zone×month; zone×hour; site-day×hour). In every stratum with both populations the ensemble MSD of the solo flights S (and of the same flights alone A) is *reweighted to the mix of the group G over the strata*: S then flies, in the statistic, where and when G flies. We compare $H_{\mathrm{loc}}(\tau)$ over **every lag** and the fitted $H$. The share of G with a counterpart in its stratum is reported; "as is" is the same comparison without the reweighting, on the same strata and units (a flight counts once per cell it crosses).

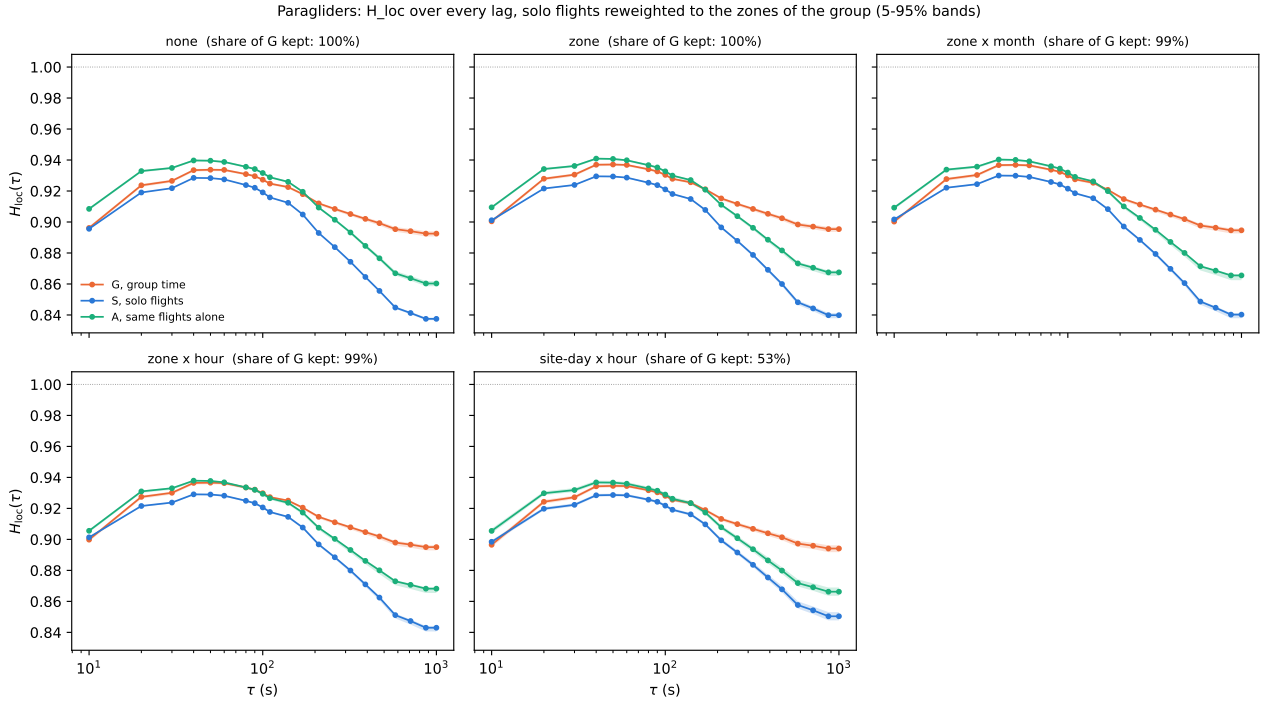



**Figure 2:** $H_{\mathrm{loc}}(\tau)$ of G (orange), of S reweighted to the strata of G (blue) and of A reweighted to them (green), 5–95% bands. First panel: all flights pooled (the raw comparison). "Share of G kept": flights of G with a solo counterpart in their stratum.

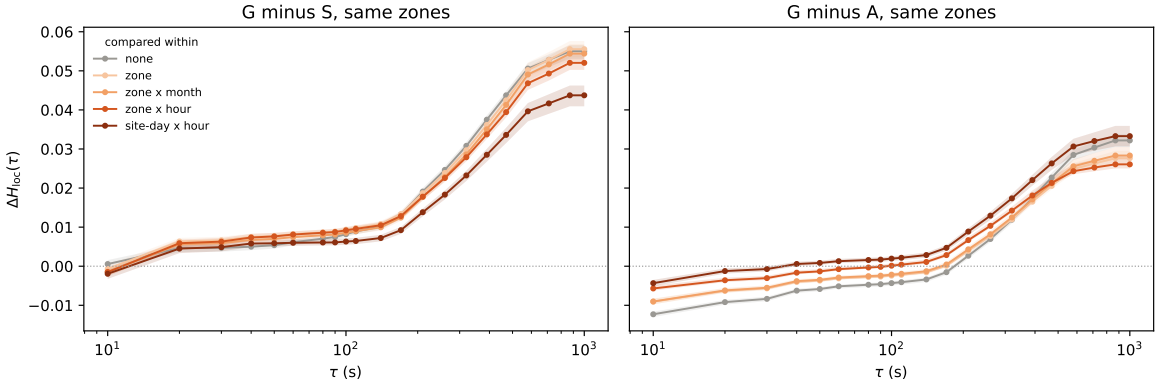



**Figure 3:** $\Delta H_{\mathrm{loc}}(\tau)$ over every lag: G minus S (left) and G minus A (right), the other population reweighted to the strata of G; one line per stratification, finer strata in darker orange.

| compared within | G kept (%) | fitted $H$ (10–1000 s) | | | $\Delta H$ | | | $\Delta H_{\mathrm{loc}}(1000\ \mathrm{s})$ | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | G | S \| G | A \| G | G − S as is | G − S reweighted | G − A reweighted | G − S as is | G − S reweighted | G − A reweighted |
| none | 100 | 0.916 | 0.899 | 0.914 | +0.017 | +0.017 [+0.016, +0.018] | +0.002 [+0.001, +0.003] | +0.055 | +0.055 [+0.053, +0.057] | +0.032 [+0.030, +0.034] |
| zone | 100 | 0.919 | 0.902 | 0.916 | +0.021 | +0.018 [+0.017, +0.019] | +0.003 [+0.003, +0.004] | +0.059 | +0.056 [+0.054, +0.058] | +0.028 [+0.026, +0.029] |
| zone × month | 99 | 0.919 | 0.902 | 0.915 | +0.022 | +0.017 [+0.016, +0.018] | +0.003 [+0.003, +0.004] | +0.064 | +0.054 [+0.053, +0.057] | +0.028 [+0.027, +0.031] |
| zone × hour | 99 | 0.919 | 0.902 | 0.913 | +0.022 | +0.017 [+0.016, +0.018] | +0.005 [+0.005, +0.006] | +0.064 | +0.052 [+0.050, +0.054] | +0.026 [+0.025, +0.028] |
| site-day × hour | 53 | 0.917 | 0.904 | 0.913 | +0.015 | +0.013 [+0.012, +0.014] | +0.008 [+0.007, +0.009] | +0.051 | +0.044 [+0.041, +0.046] | +0.033 [+0.031, +0.036] |

**Table 2:** Fitted $H$ and differences by stratification. "Reweighted": the other population reweighted to the strata of G (value, 5–95% band); "as is": the same strata without reweighting. $H_{\mathrm{S|G}}$: $H$ of S reweighted to G.

**Reading.**

- **The gap opens above $\tau \approx 150$ s.** $\Delta H_{\mathrm{loc}}(\tau)$ between G and S is small below it (+0.005 to +0.011 from 20 to 140 s) and rises to +0.055 at 1000 s, above one climb–glide cycle: the group acts on how cycles are chained, not on a single glide or climb.
- **The zone does not explain it.** Reweighting S to the cells where G flies lowers $\Delta H_{\mathrm{loc}}(1000)$ by only 6% (+0.059 → +0.056) and the fitted $\Delta H$ by 15%; with the month, 15% and 23%; with the hour, 19% and 24%. At least 80% of the long-lag gap remains. Solo pilots fly where groups fly: 100% of G has a solo counterpart in its cell.
- **Same day, site and hour** (the finest strata, 53% of G kept): $\Delta H_{\mathrm{loc}}(1000)$ =+0.044 [+0.041, +0.046], 14% below its as-is value; fitted $\Delta H$ =+0.013 [+0.012, +0.014]. Solo flights on the same day, site and hour as a group still progress less straight over 1000 s.
- **Fitted $H$:** G 0.916, S 0.899 raw. S reweighted to the zones of G: 0.902; to zone×hour: 0.902; to site-day×hour: 0.904 (G 0.917). Reweighting raises the fitted $H$ of S by a few thousandths and never brings it to that of G.
- **Against the same flights alone** (A): $\Delta H_{\mathrm{loc}}(1000)$ is +0.032 pooled and +0.028, +0.026, +0.033 in zone, zone×hour, site-day×hour strata: the same pilots in the same zones and hours progress straighter in a group. Below 150 s the sign depends on the hour: A is above G pooled and equal at the same site-day and hour, which is why the within-flight estimate of Section 3 must be read with the same-site-day-hour one.
- **Answer:** the difference between group and solo flights does not come from the zones where groups fly, nor from the month or hour at that place; it is reduced by about a fifth by the day, site and hour together, and the rest stays with the pilot and the day fixed. It is not a zone effect.

**Limits.** Cells are 25 km, from the middle of a piece; the terrain inside a cell (ridge, plain) is not controlled, and a group may choose better air within it: this is part of what we call the group effect. Paragliders only.

# 3 Other statistics: what the group changes

| | solo flights S | group time G | same flights alone A |
|---|---|---|---|
| flights (TA-MSD) | 119 129 | 33 435 | 29 611 |
| $H$ (10–1000 s) | 0.899 [0.898, 0.899] | 0.916 [0.915, 0.917] | 0.914 [0.913, 0.915] |
| $H_{\mathrm{loc}}(100\ \mathrm{s})$ | 0.919 [0.919, 0.919] | 0.927 [0.927, 0.928] | 0.932 [0.931, 0.932] |
| $H_{\mathrm{loc}}(1000\ \mathrm{s})$ | 0.837 [0.837, 0.838] | 0.893 [0.891, 0.894] | 0.860 [0.859, 0.862] |
| time in climb (%) | 30.8 [30.7, 30.9] | 33.0 [32.8, 33.2] | 29.1 [28.9, 29.2] |
| searches ending in a climb (%) | 43.3 [43.2, 43.4] | 49.5 [49.2, 49.7] | 45.9 [45.7, 46.1] |
| climb rate (m/s) | 1.22 [1.22, 1.22] | 1.26 [1.25, 1.27] | 1.33 [1.32, 1.34] |
| glide speed (km/h) | 30.1 [30.1, 30.2] | 33.3 [33.1, 33.5] | 32.8 [32.6, 32.9] |
| glide ratio | 10.4 [10.4, 10.4] | 12.9 [12.8, 13.0] | 11.3 [11.2, 11.4] |
| cross-country speed (km/h) | 17.2 [17.2, 17.3] | 20.0 [19.8, 20.3] | 20.0 [19.8, 20.1] |

**Table 3:** Pooled statistics of S, G and A (paragliders), with 5–95% bands (blocks of the note). TA-MSD on loggers of at most 10 s, phases on 1 Hz flights.

**Design.** Five contrasts, each computed inside strata and averaged over those with both populations (weights $n_P n_Q/(n_P + n_Q)$, $n$ the flights of each side): G − S pooled; G − S and A − S (*selection*) at the same site-day and hour; G − A within the flight and at the same site-day and hour (the *group effect*, estimated in two ways that fail in opposite ways: the first compares a flight with itself at another time of day, the second compares pilots at the same time). Where the last two agree, the effect belongs to the group.

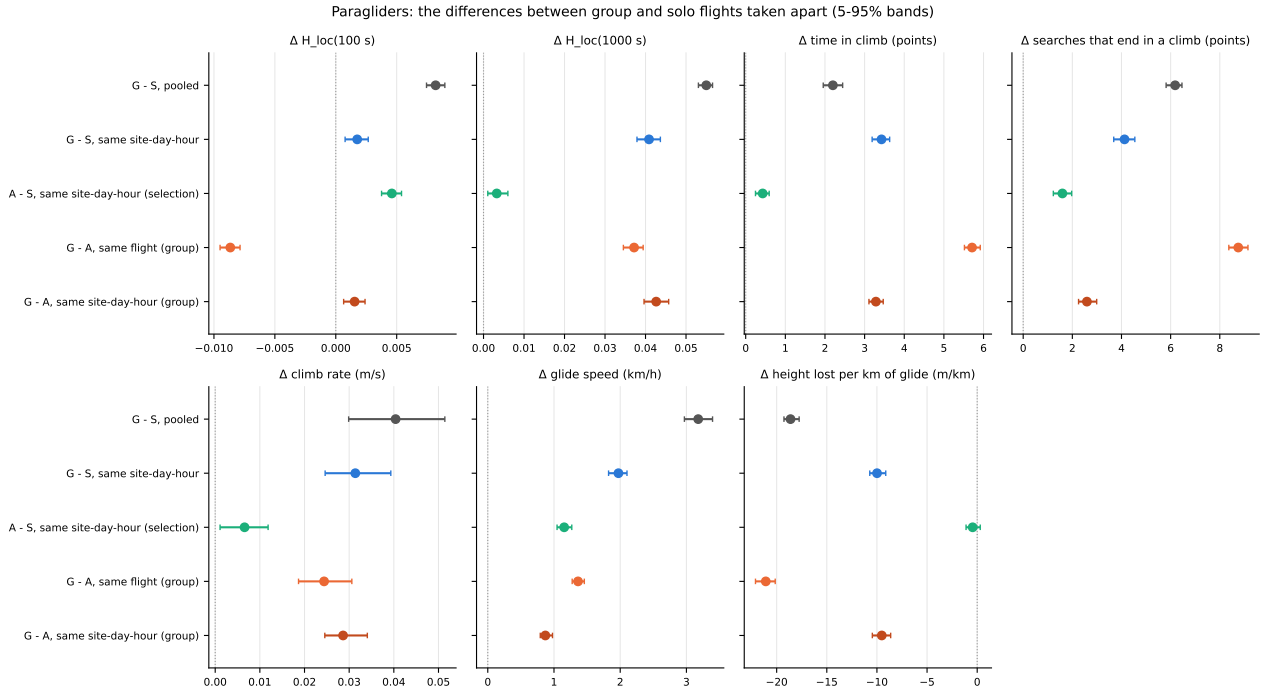



**Figure 4:** Each statistic (panel) for the five contrasts (rows), with 5–95% bands. Shares of time in points; lower height lost per km is a better glide.

| Δ | G − S pooled | G − S same s-d-h | A − S same s-d-h | G − A same flight | G − A same s-d-h |
|---|---|---|---|---|---|
| $H_{\text{loc}}(100\text{ s})$ | +0.008 [+0.007, +0.009] | +0.002 [+0.001, +0.003] | +0.005 [+0.004, +0.005] | −0.009 [−0.009, −0.008] | +0.002 [+0.001, +0.002] |
| $H_{\text{loc}}(1000\text{ s})$ | +0.055 [+0.053, +0.057] | +0.041 [+0.038, +0.044] | +0.003 [+0.001, +0.006] | +0.037 [+0.035, +0.039] | +0.043 [+0.040, +0.046] |
| time in climb (%) | +2.20 [+1.96, +2.45] | +3.43 [+3.19, +3.63] | +0.43 [+0.25, +0.59] | +5.71 [+5.52, +5.92] | +3.28 [+3.11, +3.47] |
| searches to climb (%) | +6.18 [+5.82, +6.46] | +4.12 [+3.68, +4.54] | +1.60 [+1.23, +1.97] | +8.75 [+8.37, +9.15] | +2.59 [+2.25, +2.99] |
| climb rate (m/s) | +0.040 [+0.030, +0.051] | +0.031 [+0.025, +0.039] | +0.007 [+0.001, +0.012] | +0.024 [+0.019, +0.031] | +0.029 [+0.025, +0.034] |
| glide speed (km/h) | +3.18 [+2.97, +3.39] | +1.97 [+1.82, +2.10] | +1.15 [+1.05, +1.27] | +1.36 [+1.27, +1.46] | +0.87 [+0.79, +0.98] |
| height lost per km (m/km) | −18.6 [−19.3, −17.8] | −10.0 [−10.7, −9.1] | −0.5 [−1.1, +0.3] | −21.1 [−22.1, −20.1] | −9.5 [−10.5, −8.6] |
| cross-country speed (km/h) | +2.81 [+2.59, +3.01] | +2.25 [+2.09, +2.39] | +1.31 [+1.21, +1.43] | +0.70 [+0.60, +0.78] | +1.29 [+1.16, +1.41] |

**Table 4:** The contrasts of Figure 4: difference and 5–95% band (s-d-h: same site-day and hour).

**Reading.**

- **Straighter progress over 1000 s** survives every control: $\Delta H_{\text{loc}}(1000)$ is +0.055 pooled, +0.037 within the flight, +0.043 at the same site-day and hour; selection only +0.003.
- **Nothing within one phase (100 s):** −0.009 within the flight, +0.002 at the same place and hour; the pooled +0.008 was selection and composition.
- **More time climbing** (+5.71, +3.28 points) and **more searches that end in a climb** (+8.75, +2.59 points).
- **Climb rate is higher in a group** (+0.024, +0.029 m/s), not lower as in the pooled table: a composition effect (pooled ratios weight flights by their time in each population).
- **Faster and flatter glides:** +1.36, +0.87 km/h and −21.1, −9.5 m/km less height lost per km; selection adds +1.15 km/h of glide speed (group pilots glide faster even alone).
- **Cross-country speed** is higher in a group once stratified: +0.70, +1.29 km/h.

## 3.1 Size of the configuration

The group time is cut where the configuration changes: every stretch has a fixed number of pilots $n$ (2, 3, 4, 5–6, 7+); a flight can contribute to several sizes. Two contrasts by size: each size minus the same flights alone (within the flight), and each size minus configurations of 2 pilots (same site-day and hour: does the size matter beyond a pair?).

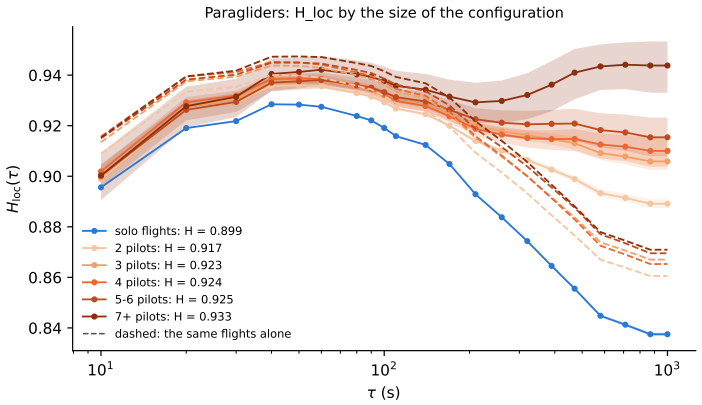



**Figure 5:** $H_{\text{loc}}(\tau)$ over every lag by the size of the configuration (orange, darker with more pilots), S (blue) and, dashed, the same flights alone for each size.

| | | $H$ | | $H_{\text{loc}}(1000\text{ s})$ | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| pilots | flights | $G_n$ | alone | $G_n$ | alone | time in climb (%) | searches to climb (%) | climb rate (m/s) | glide speed (km/h) | glide ratio |
| solo | 119 129 | 0.899 | | 0.837 | | 30.8 | 43.3 | 1.22 | 30.1 | 10.4 |
| 2 | 26 411 | 0.917 | 0.914 | 0.889 | 0.861 | 32.7 | 49.0 | 1.27 | 33.2 | 12.5 |
| 3 | 9 480 | 0.923 | 0.919 | 0.906 | 0.867 | 33.6 | 49.9 | 1.26 | 33.5 | 13.1 |
| 4 | 4 073 | 0.924 | 0.920 | 0.910 | 0.865 | 33.6 | 51.0 | 1.25 | 33.5 | 13.7 |
| 5–6 | 2 686 | 0.925 | 0.922 | 0.915 | 0.870 | 33.1 | 50.1 | 1.23 | 33.1 | 14.1 |
| 7+ | 1 142 | 0.933 | 0.924 | 0.944 | 0.871 | 34.6 | 51.7 | 1.20 | 34.6 | 15.9 |

**Table 5:** Paragliders, pooled by the size of the configuration ("alone": the same flights alone).

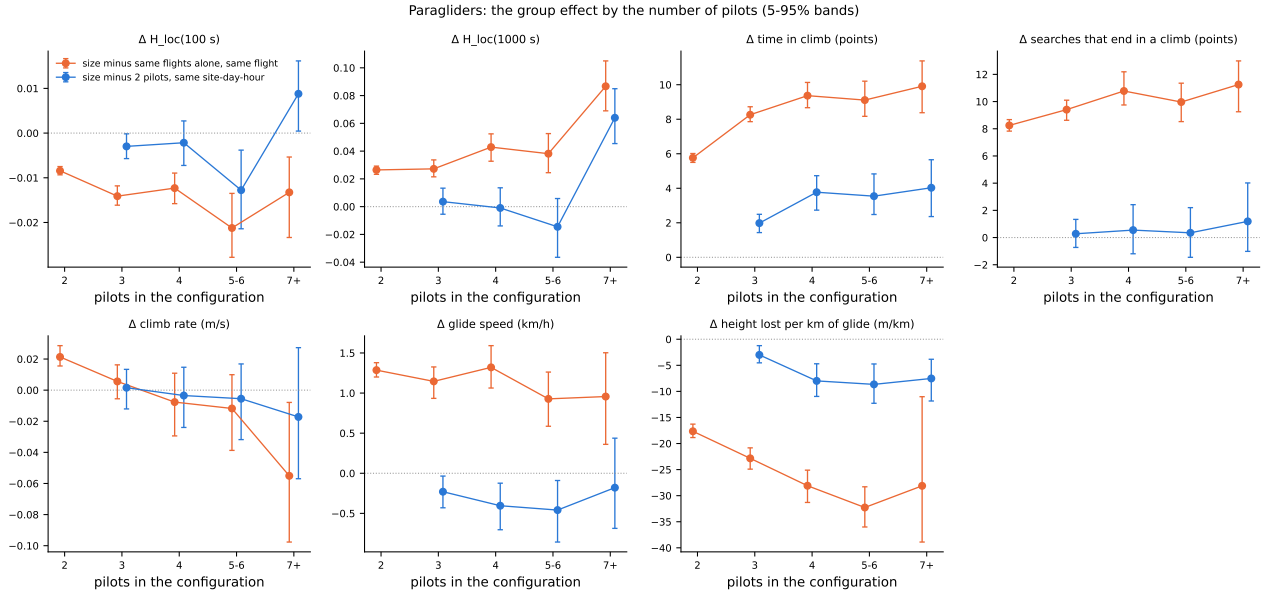



**Figure 6:** Orange: $G_n - A$ within the flight. Blue: $G_n - G_2$ at the same site-day and hour.

**Reading.** Pooled, $H$ and $H_{\text{loc}}$ rise with $n$ while the same flights alone stay flat (Fig. 5, Table 5). Within the flight the straight-progress effect grows with $n$ (+0.026 with 2 pilots, +0.087 with 7+), but at the same site-day and hour configurations of 3 to 6 pilots do not differ from pairs ($\Delta H_{\text{loc}}(1000)$ between −0.015 and +0.004, bands across zero): the rise of the pooled curves belongs to the days and sites where large groups form. Only 7+ stands apart (+0.064 [+0.045, +0.085], cross-country speed +7.0 km/h), a rare configuration (3% of group time) of unknown nature. Time in climb and the flatter glide do grow with $n$ at the same place and hour (+2.0 to +4.0 points; −3 to −9 m/km).

## 3.2 How groups form and break, who leads, when they fly

- **Entering and leaving.** The phase at the instant a flight enters (31 381) or leaves (31 202) a group, against the time shares; entries and exits at the departure or the landing apart (1 179, 1 358).

- **Leader and follower.** In configurations of two pilots the signed median delay of the pair interval says who passes first (leader: the other passes at least 15 s later; 35% abreast, median delay 20 s). The two are compared within the configuration.
- **Habitual leaders.** For the 946 pilots with at least 10 roles in pairs: dispersion of the lead fractions against roles swapped at random.
- **Hours.** Time of S, A and G over the local hour.

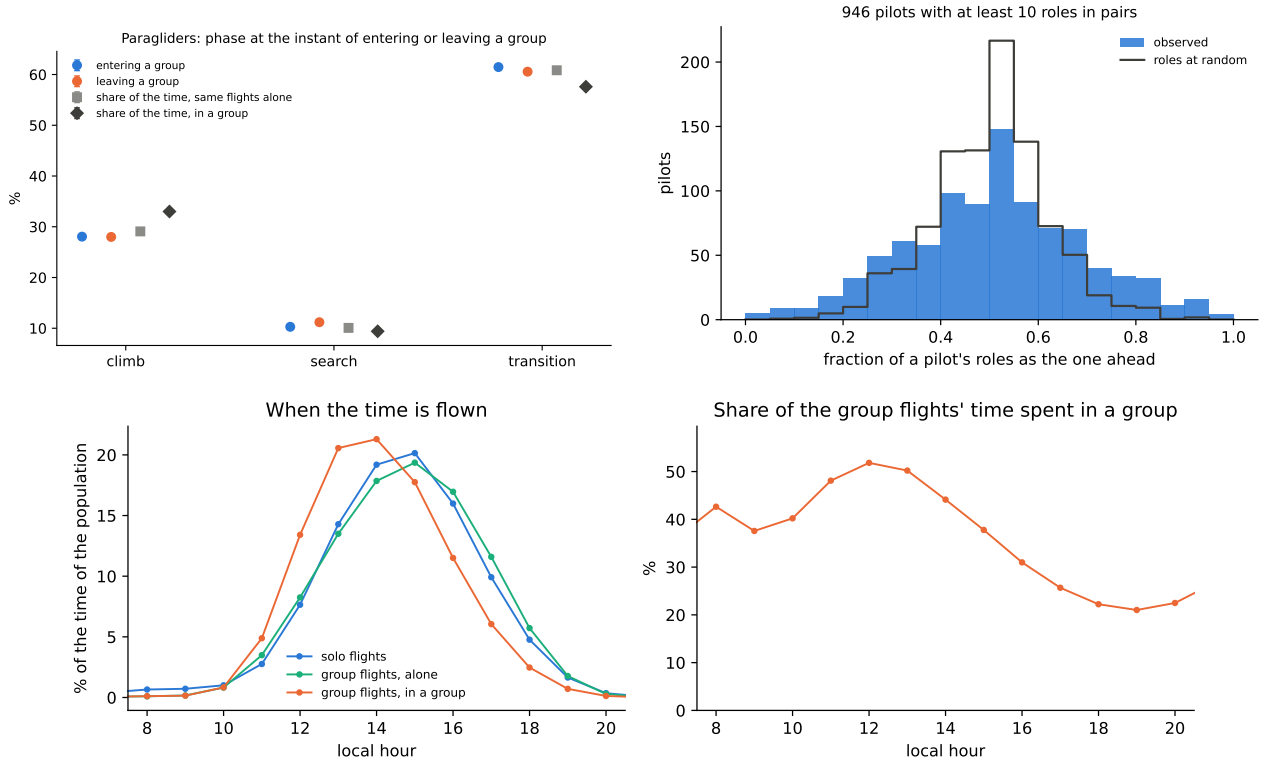



**Figure 7:** Top left: phase at entry and exit against the time shares (bands: blocks). Top right: lead fraction of each pilot, against roles at random. Bottom: time over the local hour (left) and share spent in a group (right).

**Reading.** Entries fall in climbs 28.1%, searches 10.3% and glides 61.5% of the cases, exits 28.0, 11.2, 60.6%, against 29.1, 10.1, 60.8% of the time alone: **no phase where groups preferentially form or break**. 34% of entries are within 10 min of the departure. The pilot ahead flies a little faster (glide speed $+0.86$ [$+0.74$, $+0.99$] km/h, climb rate $+0.021$ [$+0.014$, $+0.028$] m/s) but does not find thermals more often (searches ending in a climb $-0.01$ [$-1.03$, $+0.83$] points, height lost per km $+1.6$ [$-0.9$, $+4.5$] m/km): within a pair the follower shows no sign of using the leader to find the next thermal. Dispersion of lead fractions 2.73 against 1.00 at random (95% below 1.07): **some pilots lead or follow persistently** (12% lead in at most 20% or at least 80% of their roles, 3% expected). Group time is concentrated around midday (52% of a group flight's time at 12 h, 38% at 9 h, 22% at 18 h), and its time alone falls earlier and later: this is why G $-$ A within the flight must be read with the same-site-day-hour estimate.

# 4 Conclusions and limits

**Supported**

- the progress over a quarter of an hour is straighter in a group, with the pilot, the day, the site and the hour fixed;
- no effect within one glide or climb (100 s) at the same hour;
- more time climbing, more searches that end in a climb, faster and flatter glides;
- climb rate and cross-country speed are *higher* in a group once the composition is controlled (the pooled readings reversed);
- selection of pilots is small for the motion;
- groups form and break in every phase; some pilots lead persistently.

**Not supported**

- that the group *causes* the effect: the strata fix pilot, day, site, hour and zone, not the decision to fly together;
- that larger groups straighten the progress, up to six pilots;
- that a contact means interaction (shared air within sight);
- the glide statistics are partly by definition (aligned headings in glide);
- phases exist on 1 Hz flights only; the same analyses on the xc-dynamics tracks are open.

# A How $H$ and $H_{\mathrm{loc}}$ are computed

$\mathrm{MSD}(\tau) \propto \tau^{2H}$, so $\log_{10}\mathrm{MSD} = 2H\log_{10}\tau + c$. $H$ is half the slope of the least-squares line over all 21 lags (equal weight per lag). $H_{\mathrm{loc}}(\tau_c)$ is half the slope of the line fitted on the lags with $|\log_{10}\tau - \log_{10}\tau_c| \leq 0.25$ (a factor 1.78 on each side), equal weight; where fewer than 3 lags lie in the window, the 3 nearest are used. It is a windowed regression, not a pointwise derivative. $\Delta H$ and $\Delta H_{\mathrm{loc}}$ of two curves are the differences of the fitted values; for a stratified average they are the same regression on the averaged $\Delta\log\mathrm{MSD}$ (linear: slope of a difference = difference of slopes). Bands: the same computation on each bootstrap draw, percentiles lag by lag.

| $\tau$ (s) | lags used | $\tau$ (s) | lags used | $\tau$ (s) | lags used |
|---|---|---|---|---|---|
| 10 | 10–30 (3 nearest) | 90 | 60–140 | 320 | 210–470 |
| 20 | 20–40 (3 nearest) | 100 | 60–170 | 390 | 260–580 |
| 30 | 20–50 | 110 | 80–170 | 470 | 320–710 |
| 40 | 30–60 | 140 | 80–210 | 580 | 390–1000 |
| 50 | 30–80 | 170 | 100–260 | 710 | 470–1000 |
| 60 | 40–100 | 210 | 140–320 | 870 | 580–1000 |
| 80 | 50–140 | 260 | 170–390 | 1000 | 580–1000 |

**Table 6:** Lags of the window of each $H_{\mathrm{loc}}$ value. At 10 and 20 s the window is not centred; at 870 and 1000 s it is cut at 1000 s and the two values use the same lags, so the end of the curve flattens by construction. The window resolves nothing narrower than about half a decade.

**Phase MSD figures (not in this report)** use a different local exponent: the finite-difference derivative of $\log\mathrm{MSD}$ against $\log\Delta$ on 100 lags, without the factor $1/2$ (about 2 for ballistic motion).

**Ratios and the glide ratio.** Ratios are sums of numerators over sums of denominators over the runs of a population (a run goes to the stretch that holds its middle). In a small stratum a glide in rising air has almost no height lost, the glide ratio diverges and its average over strata is unstable; stratified contrasts use the *height lost per km of glide*, the inverse. Pooled tables keep the glide ratio.

# B Checks of the grouping and of the viewer

- **Code against the methodological note.** Every phase was compared with its code (grid, departure, candidate filter, scan of delays, contact, gaps, signed delay, null model, configurations, episodes, membership, blocks): all implemented as written, with four differences of text, **now corrected in red in the note**: (1) the phases of the run are Vilpellet's, on 1 Hz flights, not the hidden Markov model of xc-lab, so the heading condition acts only between two such flights; (2) headings are in the east and north of the first fix of each flight, not of the fix itself (a rotation of about 3° at 300 km, against 45°); (3) a contact is kept only from the later departure to the earlier landing (at most $\delta$ lost at the ends); (4) instants in a gap of the track are labelled solo in the tables.
- **Properties of the tables** (chosen run): every pair interval lasts at least $\tau^* = 2285$ s and lies where both flights fly after departure; every configuration has at least two members; membership covers each flight from takeoff to landing without holes; on 3 000 flights the group time equals the union of its pair intervals; on 3 000 configurations the members are connected at the midpoint and no active interval leaves; on 2 000 episodes every configuration but the first starts when another ends and shares a member; every episode lies in one block. The counts of the note (154 598 and 6 039 flights; 21.7% and 6.9% in a group) are reproduced.
- **Viewer.** The *Group flights* tab reads `groups/` of its data folder: its seven tables are **byte-identical** to those of the chosen run; the tracks it draws (`SSD_DATA`) have the same fixes, times, positions and trimmed starts as those the grouping read (`SSD_DISANTE`).
- **Analyses.** Tests on synthetic flights for every step, and on every number of the README of the analyses; the decomposition is checked on a synthetic case where the day raises a statistic by 10 and the group by 2: pooled 10, stratified 2.

# C Which statistics are drawn

Unit: a population on a site-day with at least 25 flights (119 units). A statistic is redundant when the others explain most of it across units ($R^2$ of its linear regression on them); backward elimination drops the most explained until every $R^2 \leq 0.5$. The four shares of time sum to 100%, so one of them goes first.

| statistic | status | $R^2$ on the five drawn |
|---|---|---|
| time in climb (%) | drawn | 0.23 |
| searches that end in a climb (%) | drawn | 0.41 |
| climb rate (m/s) | drawn | 0.20 |
| glide speed (km/h) | drawn | 0.26 |
| glide ratio | drawn | 0.38 |
| time in transition (%) | dropped, number 1, $R^2$ 1.00 when dropped | 0.83 |
| height gained per climb (m) | dropped, number 2, $R^2$ 0.98 when dropped | 0.91 |
| climbs that start after a search (%) | dropped, number 3, $R^2$ 0.93 when dropped | 0.77 |
| cross-country speed over 30 min (km/h) | dropped, number 4, $R^2$ 0.93 when dropped | 0.90 |
| time in a search in vain (%) | dropped, number 5, $R^2$ 0.88 when dropped | 0.71 |
| climbs per hour | dropped, number 6, $R^2$ 0.72 when dropped | 0.71 |
| time in a search that ends in a climb (%) | dropped, number 7, $R^2$ 0.65 when dropped | 0.65 |

**Table 7:** Five statistics remain; the others are explained by them.

# D Pooled curves, hang gliders and sizes

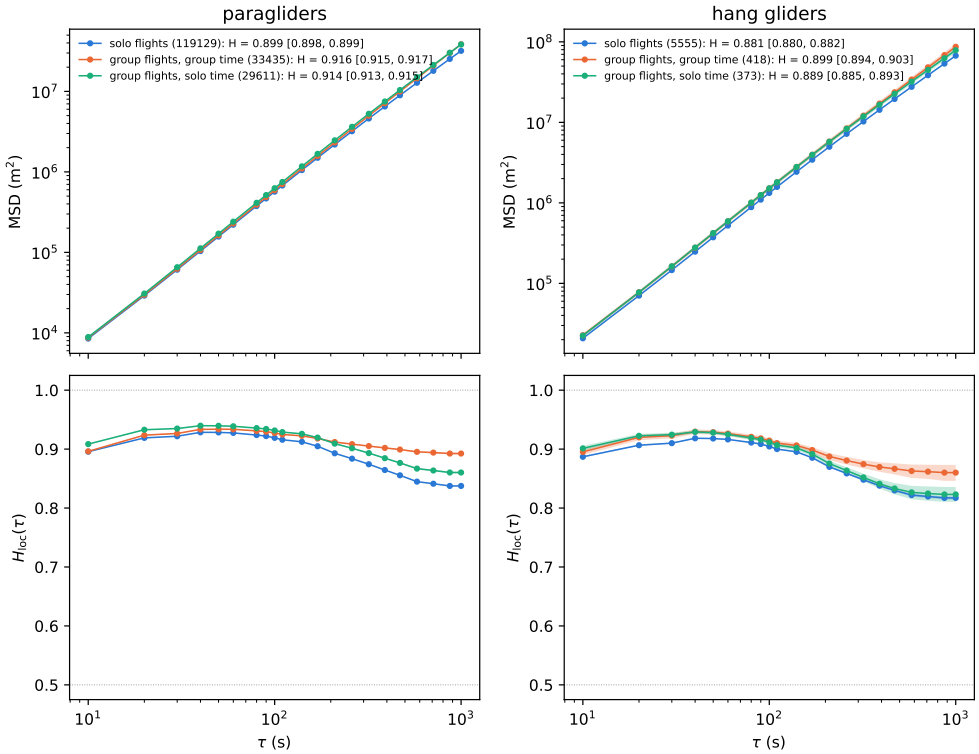



**Figure 8:** TA-MSD and $H_{loc}$ of S, G and A, paragliders and hang gliders (hang gliders: 418 group flights, wide bands at long lags).

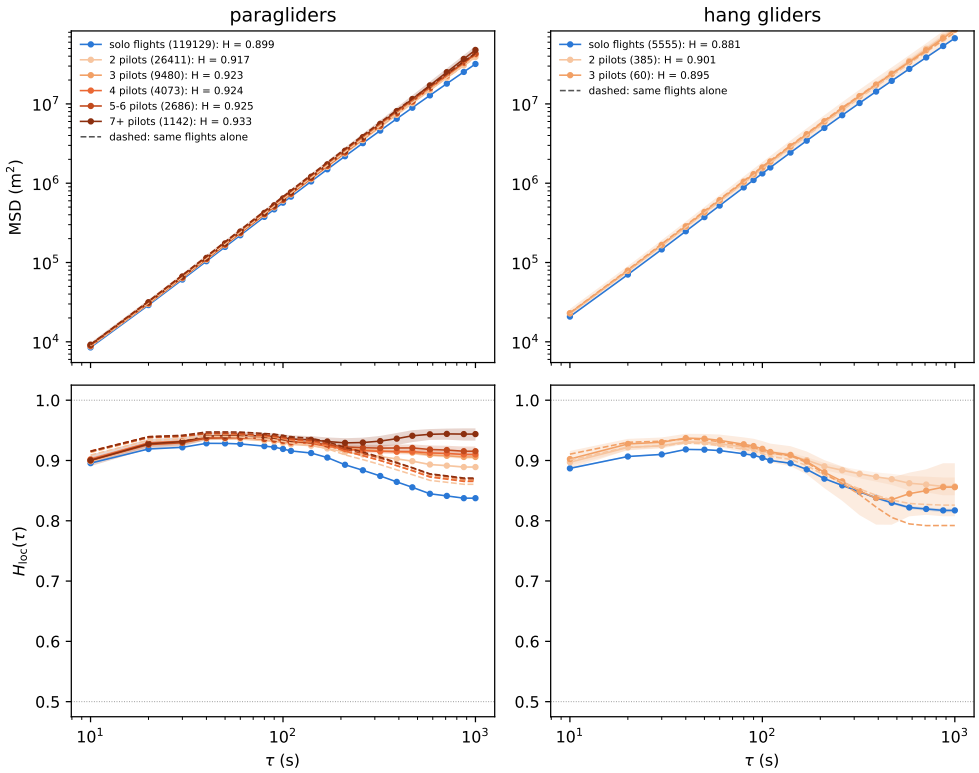



**Figure 9:** By the size of the configuration (orange, darker with more pilots), S (blue) and, dashed, the same flights alone for each size.

# E Single site-days

Six site-days chosen by a fixed rule (most sizes with 15 flights or more, then most flights, at most two per takeoff site). Bands resample flights (too narrow: flights of one group are not independent).

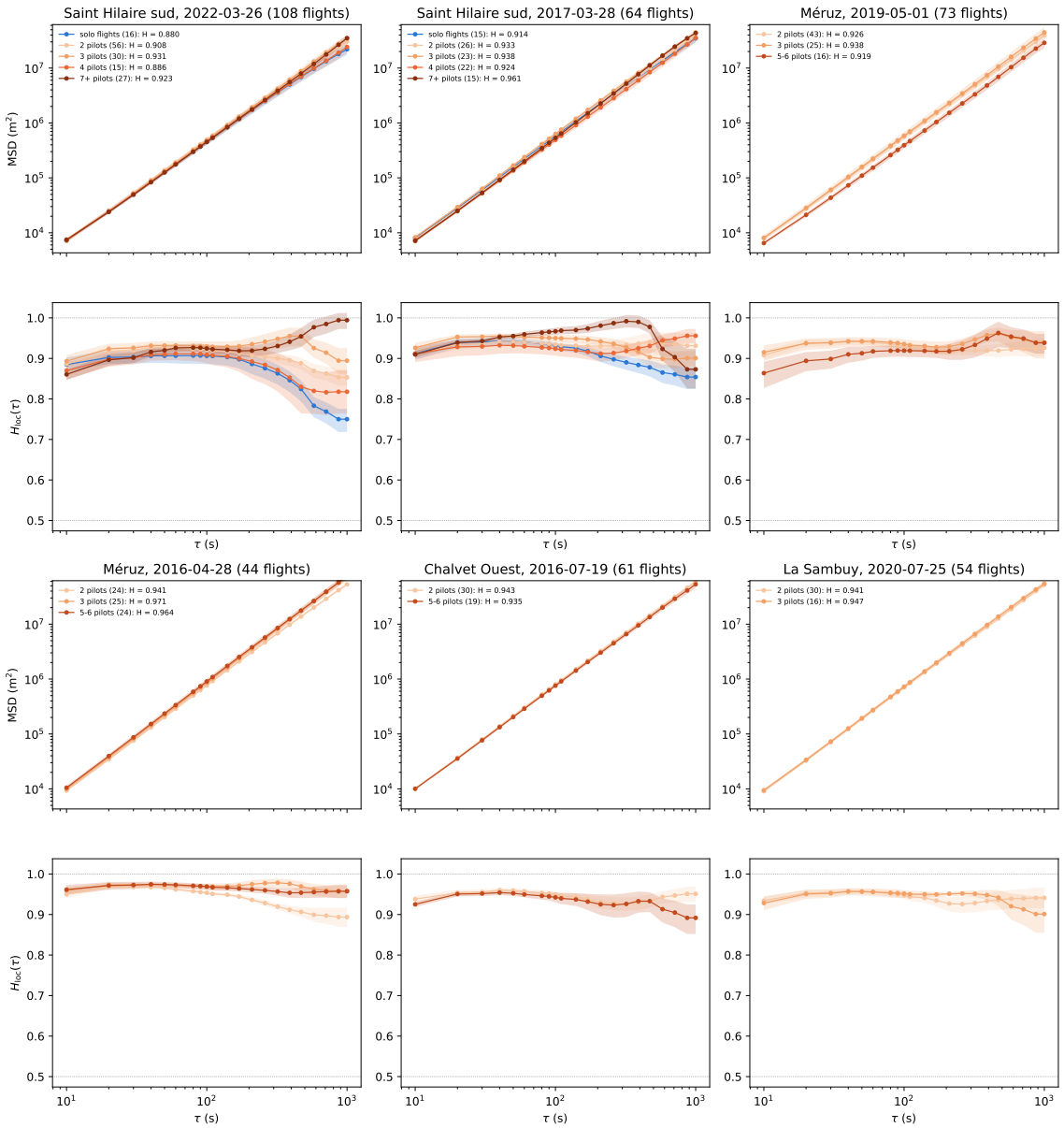



**Figure 10:** TA-MSD and $H_{\text{loc}}$ on six site-days by the size of the configuration, with the solo flights of the same site-day where they reach 15 flights.

| site-day | solo | 2 | 3 | 4 | 5–6 | 7+ |
|---|---|---|---|---|---|---|
| Saint Hilaire sud, 2022-03-26 | 0.750 | 0.853 | 0.894 | 0.818 | | 0.994 |
| Saint Hilaire sud, 2017-03-28 | 0.854 | 0.931 | 0.901 | 0.956 | | 0.873 |
| Méruz, 2019-05-01 | | 0.926 | 0.939 | | 0.938 | |
| Méruz, 2016-04-28 | | 0.894 | 0.958 | | 0.957 | |
| Chalvet Ouest, 2016-07-19 | | 0.951 | | | 0.892 | |
| La Sambuy, 2020-07-25 | | 0.941 | 0.901 | | | |

**Table 8:** $H_{\text{loc}}(1000\text{ s})$ by size on each site-day (empty: fewer than 15 flights). Differences between days exceed those between sizes (the climb rate goes from 0.79 to 2.11 m/s across the six days).

# F Robustness to the threshold

With $q = 0.2$ contacts need to last only 26.4 min: more and shorter groups, up to one contact in five chance.

| | $q = 0.1$, $\tau^* = 38.1$ min | $q = 0.2$, $\tau^* = 26.4$ min |
|---|---|---|
| group flights (TA-MSD) | 33 435 | 48 777 |
| $H_{\text{loc}}(1000\text{ s})$, S | 0.837 | 0.836 |
| $H_{\text{loc}}(1000\text{ s})$, A | 0.860 | 0.854 |
| $H_{\text{loc}}(1000\text{ s})$, G | 0.893 | 0.892 |
| $\Delta H_{\text{loc}}(1000\text{ s})$, G − A same flight | +0.037 [+0.035, +0.039] | +0.032 [+0.029, +0.035] |
| $\Delta H_{\text{loc}}(1000\text{ s})$, G − A same s-d-h | +0.043 [+0.040, +0.046] | +0.042 [+0.040, +0.044] |
| $\Delta H_{\text{loc}}(1000\text{ s})$, A − S same s-d-h | +0.003 [+0.001, +0.006] | +0.002 [−0.001, +0.005] |
| $\Delta H_{\text{loc}}(100\text{ s})$, G − A same flight | −0.009 [−0.009, −0.008] | −0.016 [−0.017, −0.015] |
| $\Delta H_{\text{loc}}(100\text{ s})$, G − A same s-d-h | +0.002 [+0.001, +0.002] | −0.004 [−0.005, −0.003] |
| Δ searches to climb, G − A same s-d-h (points) | +2.6 [+2.3, +3.0] | +3.4 [+3.0, +3.7] |
| Δ climb rate, G − A same s-d-h (m/s) | +0.029 [+0.025, +0.034] | +0.024 [+0.020, +0.029] |
| Δ height lost per km, G − A same s-d-h (m/km) | −9.5 [−10.5, −8.6] | −11.5 [−12.2, −10.9] |
| Δ cross-country speed, G − A same s-d-h (km/h) | +1.29 [+1.16, +1.41] | +1.14 [+1.01, +1.30] |

**Table 9:** The main results with the two thresholds (s-d-h: same site-day and hour). Every conclusion keeps its sign; at 100 s both estimates of the group effect are slightly negative with $q = 0.2$: no robust effect within one phase.

# G Leader minus follower

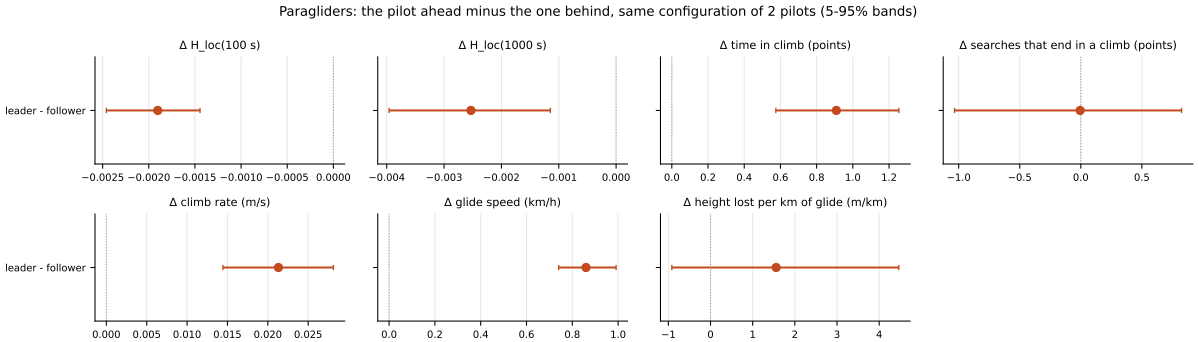



**Figure 11:** Leader minus follower, same configuration of two pilots.

| Δ | leader − follower |
|---|---|
| $H_{\text{loc}}(100\text{ s})$ | −0.002 [−0.002, −0.001] |
| $H_{\text{loc}}(1000\text{ s})$ | −0.003 [−0.004, −0.001] |
| time in climb (%) | +0.91 [+0.57, +1.25] |
| searches ending in a climb (%) | −0.01 [−1.03, +0.83] |
| climb rate (m/s) | +0.021 [+0.014, +0.028] |
| glide speed (km/h) | +0.86 [+0.74, +0.99] |
| height lost per km of glide (m/km) | +1.6 [−0.9, +4.5] |
| cross-country speed (km/h) | +0.14 [+0.11, +0.16] |

**Table 10:** Leader minus follower, same configuration of two pilots (difference with 5–95% band).

| | climb (%) | search (%) | transition (%) |
|---|---|---|---|
| entering a group | 28.1 [27.6, 28.5] | 10.3 [10.0, 10.6] | 61.5 [61.0, 61.9] |
| leaving a group | 28.0 [27.6, 28.3] | 11.2 [10.9, 11.5] | 60.6 [60.1, 61.0] |
| share of time, alone | 29.1 [28.9, 29.3] | 10.1 [10.0, 10.2] | 60.8 [60.6, 61.0] |
| share of time, in a group | 33.0 [32.8, 33.2] | 9.4 [9.2, 9.6] | 57.6 [57.3, 57.9] |

**Table 11:** Phase at the instant of entering or leaving a group, and shares of time in each phase (percent, with 5–95% bands).

# H Reproducing the report

From `xc-lab/analyses/groups`, SSD_DISANTE mounted:

```
uv run python tamsd.py; uv run python phase_msd.py; uv run python survival.py
uv run python budget.py; uv run python performance.py; uv run python blocks.py
uv run python redundancy.py; uv run python size.py; uv run python decompose.py
uv run python zones.py; uv run python mechanisms.py; uv run python report/facts.py
uv run python report/tables.py; cd report; latexmk -pdf group_vs_solo.tex
```

Heavy pieces stay on SSD_DISANTE, in the `analyses/` folder of each run; figures and JSON in `out/<run>/`.